



Investigating the impact of immediate vs. delayed feedback timing on motivation and language learning outcomes in online education: Perspectives from Feedback Intervention Theory

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ABSTRACT

Despite extensive research on feedback in language learning, a notable gap persists regarding the influence of feedback timing—specifically immediate versus delayed—on motivation and learning outcomes in online education, particularly among Chinese EFL learners. This study addressed this gap by examining the effects of feedback timing on motivation and language learning outcomes within the context of China's online learning environment, employing Feedback Intervention Theory (FIT) as its theoretical foundation. The research involved an experimental design with 98 intermediate EFL students, randomly allocated to three groups: an immediate feedback group (IFG), a delayed feedback group (DFG), and a no-feedback group (NFG). Participants completed pretests assessing motivation and English proficiency, followed by an educational task tailored to their assigned feedback condition. Posttests measured changes in motivation and language learning outcomes, with one-way ANOVA applied to analyze differences across groups. Findings revealed that both immediate and delayed feedback significantly enhanced motivation and language learning outcomes compared to the no-feedback condition, though no substantial differences emerged between IFG and DFG. These results indicate that integrating timely feedback into online education can effectively support Chinese EFL learners' motivation and language learning outcomes.

1. Introduction

Feedback constitutes a cornerstone of the educational process, furnishing learners with critical insights into their performance while illuminating their strengths and areas requiring enhancement (Jin et al., 2025; Quinn & Nakata, 2017). Through the provision of precise and meaningful information, feedback equips students to refine their approaches, enrich their understanding, and make informed choices about their learning trajectories (Fu & Li, 2022; Greisel et al., 2025). Its significance extends beyond mere error correction, as it reshapes learners' cognitive frameworks and bolsters their capacity to apply knowledge across diverse contexts (Kennedy et al., 2025). Far from solely addressing deficiencies, feedback also celebrates achievements and reinforces competencies, thereby elevating learners' confidence and sustaining their drive to persist (Kinder et al., 2025; Mao & Lee, 2020). Moreover, according to McCue et al. (2025), when delivered promptly, feedback acts as a powerful motivator, guiding learners toward improvement and fostering a sense of competence vital for maintaining engagement. Constructive feedback nurtures a growth-oriented mindset, instilling the belief that abilities can evolve through dedication and study. When executed effectively, it enhances academic outcomes

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by enabling learners to master critical skills and knowledge (Henderson, 2021; Zahari et al., 2025). Beyond these cognitive benefits, feedback shapes emotional responses to tasks, mitigating frustration and cultivating a constructive attitude toward challenges, thus preparing learners for sustained educational success (Zhang et al., 2025).

In the realm of online education, the timing of feedback emerges as a pivotal factor influencing both motivation and learning outcomes. Defined as information conveyed to learners about their performance to enhance their knowledge and skills, feedback assumes heightened importance in digital learning environments (Clinton-Lisell, 2024). Prior studies comparing immediate and delayed feedback timing reveal notable impacts on student engagement, performance, and emotional well-being across varied educational settings (Feng et al., 2025; Lu et al., 2023). Immediate feedback, provided shortly after a learner's action or response, delivers timely insights that facilitate swift adjustments, thereby improving task execution and reinforcing a sense of mastery (Kim & Arbel, 2019; Muis et al., 2015). This immediacy sustains motivation by reducing frustration and encouraging persistent effort, particularly in fast-paced online contexts (Theobald & Bellhäuser, 2022). Conversely, delayed feedback, offered after a temporal interval, affords learners opportunities to reflect on their errors before receiving guidance (Heyne et al., 2025). While immediate feedback excels in fostering motivation, delayed feedback promotes self-directed learning and deeper comprehension by encouraging students to analyze their missteps independently (Hakim et al., 2022). Research suggests that this reflective process, often linked to delayed timing, fosters self-regulation and strengthens long-term retention and critical thinking (Attali & Van Der Kleij, 2017; Liu & Wang, 2023). Additionally, delayed feedback enhances learner autonomy, prompting independent problem-solving and reflective practices that are invaluable in online education (Clinton-Lisell, 2024).

The dynamics of immediate versus delayed feedback timing can be further elucidated through the lens of Feedback Intervention Theory (FIT), which posits that feedback influences performance by directing attention to task-specific processes, goals, and self-regulation (Alder, 2007; Li, Huang, et al., 2025; Li, Ou, et al., 2025). According to FIT, immediate feedback aligns with task-focused attention, swiftly correcting errors and reinforcing correct responses, which heightens motivation and immediate learning gains (Kim & Arbel, 2019). This rapid intervention minimizes cognitive overload and sustains learner engagement by keeping attention anchored to the task at hand (Muis et al., 2015). In contrast, delayed feedback shifts focus toward meta-cognitive processes, encouraging learners to evaluate their strategies and develop self-regulatory skills over time (King, 2016; Theobald & Bellhäuser, 2022). By allowing space for reflection before feedback is provided, according to Liu and Wang (2023), this approach fosters deeper processing and strengthens learners' ability to internalize and transfer knowledge. FIT thus underscores how timing modulates feedback's effectiveness: immediate feedback drives short-term performance and motivation, while delayed feedback cultivates enduring understanding and independence (Alder, 2007; Kluger & DeNisi, 1996). Together, these insights highlight the nuanced interplay between feedback timing, learner cognition, and educational outcomes in online settings.

The influence of feedback timing extends beyond measurable learning gains to encompass the emotional and motivational dimensions of the educational process. Motivation, defined as the psychological impetus that sparks and sustains learners' interest in tasks (Alfares, 2025; Beltman & Poulton, 2025), plays a critical role in determining how feedback is perceived and utilized. Engaged and motivated learners are more inclined to process feedback effectively, internalize linguistic knowledge, and improve their performance (Bardach & Murayama, 2024). This motivation may stem from intrinsic sources, such as a genuine interest in language mastery, or extrinsic drivers, like achieving high marks (Pat-El et al., 2024). Intrinsically motivated learners tend to view feedback as a tool for self-improvement, whereas those driven by extrinsic goals might prioritize external validation, occasionally diminishing the feedback's developmental value. Li, Huang, et al. (2025), Li, Ou, et al. (2025) and Nakata (2015) emphasize that feedback yields optimal results when its timing and tone align with learners' motivational orientations, thus amplifying its impact on their engagement.

The timing of feedback exerts a profound influence on EFL learners' motivation, a relationship that can be illuminated through the lens of FIT. According to FIT, feedback is most effective when it focuses learners' attention on task-related goals and supports their sense of progress, rather than shifting focus to self-evaluation or external distractions (Li, Huang, et al., 2025; Li, Ou, et al., 2025; Xu & Zeng, 2023). Immediate feedback, provided promptly after a task, can enhance motivation by reinforcing learners' competence and reducing uncertainty, aligning with their intrinsic desire to master language skills (Gan, 2020; Schweder & Raufelder, 2024). This timely intervention often fosters a positive emotional response, such as increased confidence, which sustains engagement in the learning process (Ni, 2024; Papi et al., 2020). Conversely, delayed feedback, while potentially disrupting this immediacy, offers unique motivational benefits when designed thoughtfully. By encouraging learners to reflect on their performance independently, it can deepen intrinsic motivation, provided sufficient guidance is available to prevent frustration (Kuklick, 2024; Pan et al., 2024). Thus, through the FIT perspective, the interplay of immediate and delayed feedback timing reveals distinct pathways to bolstering EFL learners' motivation, contingent on how each aligns with their task-oriented goals and emotional needs.

This study examined language learning outcomes as the second focal point of investigation. Language learning outcomes, for English as a Foreign Language (EFL) learners, refer to the measurable improvements in linguistic competence, including skills such as vocabulary acquisition, grammatical accuracy, and communicative proficiency (Duppenthaler, 2002; Farrokhnia et al., 2025). These outcomes are critical for EFL learners because they directly influence their ability to engage with globalized contexts, access educational opportunities, and achieve academic and professional success. Moreover, strong language learning outcomes bolster learners' confidence and persistence, which are essential for mastering a non-native language in often challenging instructional settings (Lammey et al., 2024).

The influence of feedback timing on EFL learners' language learning outcomes can be productively explored through the lens of FIT. According to FIT, the timing of feedback affects learners' attention to the task and their subsequent performance (Henderson, 2021; Kluger & DeNisi, 1996). Immediate feedback, delivered promptly after a task, may enhance language learning outcomes by reinforcing correct responses and enabling swift error correction, thus maintaining learners' focus on linguistic accuracy (Fisher et al.,

2025; Van der Kleij et al., 2012). In contrast, delayed feedback, provided after a time lapse, could encourage deeper processing and self-reflection, potentially fostering long-term retention of language skills among EFL learners (Bielak & Mystkowska-Wiertelak, 2023; Lammey et al., 2024). However, the effectiveness of each approach likely varies depending on task complexity and learner needs, suggesting a nuanced interplay between feedback timing and language learning outcomes.

Several studies have highlighted the efficacy of immediate feedback in improving performance across various domains. For instance, Lu et al. (2023) demonstrated that providing immediate text-based feedback to foreign language learners significantly enhanced their accuracy and engagement in online assignments. Similarly, Kuklick (2024) reported that automated immediate feedback played a critical role in advancing vocabulary acquisition and grammatical accuracy among language learners. Hakim et al. (2022) further corroborated these findings by linking immediate feedback mechanisms to improved learning performance, particularly through rapid error correction. Additionally, Muis et al. (2015) underscored the broader benefits of technology-mediated immediate feedback, showing its positive impact on students' attitudes, emotions, and engagement, which collectively contributed to literacy skills development.

Another strand of research has focused on the motivational benefits of timely feedback. Fisher et al. (2025) found that formative feedback delivered promptly increased students' motivational tendencies, fostering greater engagement with learning tasks. This aligns with earlier findings by Attali and Van Der Kleij (2017), who revealed that the timing and elaboration of feedback enhanced efficiency during computer-based mathematics practice, with immediate feedback boosting student confidence. Taxipulati and Lu (2021) extended these insights by demonstrating that both the timing and content of feedback significantly influenced college students' achievement in multimedia learning environments, with timely feedback playing a crucial role in sustaining motivation. A third group of studies has sought to compare the effects of immediate and delayed feedback, offering more nuanced insights into their respective contributions. Kim and Arbel (2019) explored the role of auditory feedback in language learning contexts, finding that immediate feedback supported retention, while delayed feedback encouraged deeper reflection. Almalki and Elfeky (2022) similarly examined virtual classroom settings, reporting that both feedback modalities enhanced higher-order thinking skills. However, they noted that immediate feedback accelerated skill acquisition, whereas delayed feedback fostered sustained cognitive growth. Xie and Zhang (2024) expanded this line of inquiry by investigating peer feedback in online collaborative environments. Their findings highlighted the importance of feedback design and quality, suggesting that the timing of feedback remains a critical factor even in non-traditional instructional settings.

As can be implied from the studies-reviewed above the effects of immediate versus delayed timing feedback of EFL learners' motivation and language learning outcomes. To address this gap, this study explored the effects of immediate versus delayed timing feedback of EFL learners' motivation and language learning outcomes in China. This study holds substantial implications for EFL education in China. It underscores the potential of structured feedback to bolster motivation, a critical factor in sustaining language acquisition amid the challenges of online learning environments. By comparing the two feedback approaches, the research offers practical insights for educators seeking to optimize instructional design. While immediate feedback provides swift reinforcement, delayed feedback encourages reflection, both proving effective in this context. These findings contribute to the broader discourse on feedback efficacy, aligning with China's educational priorities of improving English proficiency through innovative, technology-supported methods. Thus, the study not only enriches the theoretical framework but also informs teaching practices tailored to the unique needs of Chinese EFL learners. The following research questions were addressed:

Research question 1: Can immediate vs. delayed feedback timing significantly promote Chinese EFL learners' motivation in online education?

Research question 1: Can immediate vs. delayed feedback timing significantly enhance Chinese EFL learners' language learning outcomes in online education?

2. Method

2.1. Design and participants

The study employed a between-subjects experimental design to examine the effects of feedback timing on motivation and learning outcomes in online EFL education. This design was chosen to allow for a direct comparison of three distinct conditions: immediate feedback, delayed feedback, and no feedback. By randomly assigning participants to these groups, the study aimed to isolate the impact of feedback timing while minimizing potential confounding variables. A between-subjects approach was deemed appropriate as it prevents carryover effects, which could occur if participants experienced multiple feedback conditions. Moreover, it allows for a clear delineation of the independent variable's influence on the dependent variables, namely motivation and learning outcomes.

Participants were recruited using stratified random sampling to ensure a representative sample of intermediate-level EFL learners. Specifically, the English language institute in Xi'an, China, was stratified based on existing class groups, and participants were randomly selected from each stratum. This sampling procedure aimed to minimize selection bias and enhance the generalizability of the findings to a broader population of EFL learners within the institute. A total of 98 intermediate-level EFL learners, aged 16–22, participated in the study. These learners, attending the institute, were selected due to their accessibility and their status as a target population for online language learning interventions. The age range was chosen to reflect typical students in this educational setting, and their intermediate proficiency level provided a consistent baseline for assessing the impact of feedback timing. Random assignment was used to divide participants into three groups: the Immediate Feedback Group (IFG, $n = 31$), the Delayed Feedback Group (DFG, $n = 33$), and the No Feedback Group (NFG, $n = 34$). To verify their language proficiency, all participants completed the Oxford Quick Placement Test (OQPT) prior to the intervention. This standardized test confirmed that they all possessed an intermediate level

of English. This step was crucial to control for initial language proficiency, thereby ensuring that any observed differences in learning outcomes could be attributed to the feedback interventions rather than pre-existing variations in language ability.

2.2. Ethical considerations

Ethical standards were rigorously upheld throughout the research process, in accordance with institutional guidelines. Before the study commenced, participants were provided with comprehensive information about the objectives, procedures, and nature of the feedback conditions. Informed consent was obtained, and participants were assured that their involvement was entirely voluntary, with the option to withdraw at any stage without penalty. All collected data were anonymized to safeguard confidentiality, and personal information was securely stored on password-protected devices. Additionally, participants were informed that the data would be used solely for research purposes. The study received ethical approval from the institutional review board of the English language institute in Xi'an, ensuring adherence to established ethical protocols.

2.3. Data collection tools

The study employed three instruments to collect data which are detailed below.

2.3.1. Oxford Quick Placement Test (OQPT)

This standardized test served as a preliminary screening tool to ascertain participants' general English proficiency. The OQPT, a widely recognized instrument, assesses grammar, vocabulary, and reading comprehension through multiple-choice questions. This test facilitated the accurate placement of participants into appropriate proficiency levels, ensuring that all participants possessed an intermediate level of English proficiency. Consequently, the OQPT contributed to the formation of a homogeneous sample, enhancing the study's internal validity by minimizing variability in baseline proficiency.

2.3.2. Researcher-Designed General English Test

A 40-item objective test was developed to measure participants' English proficiency across four domains: vocabulary, grammar, reading comprehension, and listening skills. This test was designed to align with the curriculum of the online English course, ensuring content validity. Prior to its administration, a pilot study was conducted with a small group of learners to refine item clarity and appropriateness. Expert review by experienced EFL instructors further validated the test's content relevance and construct validity. The reliability of the test was assessed using the Kuder-Richardson Formula 21 (KR-21), yielding a coefficient of .80, indicating strong internal consistency. This test was administered as both a pretest and posttest. Minor modifications were made to the posttest version to mitigate memory bias while maintaining content alignment, thereby ensuring the integrity of the longitudinal data.

2.3.3. English Learning Motivation Scale (ELMS)

Adapted from Peng (2002), the ELMS was utilized to measure participants' motivation towards learning English. This scale employs a 5-point Likert format, with higher scores reflecting greater motivation. The original 37-item scale was refined to 30 items based on factor analysis results reported by Pan and Wu (2013). Items with low factor loadings, such as "I don't like English even though I know it's important," were removed to enhance the scale's psychometric properties. The final version demonstrated excellent internal consistency, with a Cronbach's α of .92 for overall motivation, and similarly robust coefficients for subscales measuring liking, dedication, efficacy, intrinsic motivation, and extrinsic motivation.

2.4. Procedures for data analysis

This study commenced with the administration of the Oxford Quick Placement Test (OQPT) to screen participants and confirm their intermediate-level English proficiency. After this screening process, participants were randomly assigned to one of three groups: the IFG, the DFG, or the NFG. Each group then completed pretests to establish their initial levels of motivation and English proficiency. During the treatment phase, which spanned 16 one-hour sessions over eight weeks, participants engaged in structured activities tailored to their assigned feedback conditions. The IFG received real-time feedback after each task, a method designed to reinforce learning through instant correction and guidance. This group participated in a diverse range of activities, including interactive online quizzes that provided automated explanations for incorrect answers, peer review sessions where classmates exchanged written work and received instructor-annotated feedback within 24 h, and targeted grammar drills with personalized error analysis. For example, after completing a writing exercise, a participant might receive detailed feedback such as, "Your argument is well-structured, but watch for subject-verb agreement errors—here, 'the team are' should be 'the team is.'" Additionally, the IFG engaged in role-playing scenarios where verbal feedback was provided on pronunciation and fluency immediately after each dialogue. These interventions were systematically integrated into every session to ensure consistent reinforcement, allowing participants to internalize corrections and apply them in subsequent tasks.

To further enhance learning, the IFG's feedback was not only corrective but also motivational, with instructors highlighting strengths alongside areas for improvement. For instance, after a vocabulary quiz, a participant might see, "Great job using advanced terminology! For next time, try incorporating more transitional phrases to improve flow." The feedback mechanisms were also adaptive, adjusting in complexity based on individual performance—participants who struggled with certain concepts received additional examples and simplified explanations, while those who excelled were challenged with more advanced tasks. Weekly

progress summaries were emailed to each participant, consolidating feedback from all activities and suggesting personalized practice exercises. This multi-faceted approach ensured that feedback was not only immediate but also meaningful, fostering a deeper understanding of language rules and encouraging active self-correction. By maintaining this structured yet dynamic feedback system, the intervention aimed to maximize engagement and long-term retention of learning outcomes.

In contrast, the DFG experienced a structured, interval-based feedback approach designed to cultivate independent learning and self-assessment skills. Feedback was intentionally withheld until participants had completed three to four assignments—typically over a two-week cycle—within the eight-week treatment phase. This method encouraged learners to reflect on their performance and attempt self-correction before receiving external input. The DFG engaged in a variety of activities, including weekly guided reflection journals where participants analyzed their own grammatical choices, identified recurring errors, and hypothesized corrections before receiving instructor feedback. Additionally, they completed cumulative assignments, such as multi-draft essays or recorded speaking tasks, which were submitted every two weeks and returned with comprehensive, annotated feedback. For example, after submitting a series of writing exercises, a participant might receive a detailed summary like, "Your narrative coherence has strengthened, but you still tend to overuse passive voice—try revising these sentences in the active voice for greater clarity." To reinforce learning, feedback sessions were often followed by structured small-group discussions, where participants compared instructor comments, shared strategies for improvement, and collaboratively analyzed common mistakes.

The DFG's interventions were further enriched by metacognitive exercises, such as error logs where participants tracked and categorized their mistakes before receiving feedback, allowing them to recognize patterns in their language use. Instructors also provided comparative feedback, highlighting progress across assignments—for instance, "In your first draft, you struggled with article usage ('a' vs. 'the'), but your most recent submission shows much better accuracy in this area." To ensure depth of processing, feedback was not only corrective but also explanatory, with clear justifications for suggested changes (e.g., "Switching to the present perfect tense here better conveys an ongoing action"). At the midpoint and end of the eight-week phase, participants received holistic progress reports synthesizing all feedback, along with tailored practice exercises targeting persistent weaknesses. This deliberate pacing of feedback—combined with reflective tasks and peer collaboration—aimed to foster long-term retention and self-regulated learning, prioritizing thoughtful analysis over immediate correction.

The NFG received no instructional feedback beyond basic grade-based assessments throughout the same eight-week period. Participants in this group completed tasks identical to those of the IFG and DFG, including quizzes and writing exercises, but relied solely on self-assessment tools, such as answer keys provided after submission. For example, after completing a grammar test, they received only a numerical score without additional commentary, allowing them to evaluate their performance independently. This condition enabled the study to assess the impact of withheld feedback on motivation and learning outcomes in comparison to the experimental groups. At the end of the treatment phase, all three groups completed posttests identical to the pretests. These assessments measured changes in motivation and English proficiency, providing data to evaluate the effects of the feedback interventions.

2.5. Procedures for data analysis

To analyze the collected data, the authors use SPSS, version 25. Along with the descriptive statistics, a one-way ANOVA was employed. This choice was predicated on the need to simultaneously examine the influence of feedback timing (immediate, delayed, and control) on multiple dependent variables (motivation and learning outcomes). Prior to conducting the one-way ANOVA, the assumptions of normality, homogeneity of variance-covariance matrices, and independence of observations were meticulously checked. The Shapiro-Wilk test was utilized to assess normality. Results indicated that the data for each dependent variable were not significantly different from a normal distribution (Motivation: $p = 0.23$; Language learning outcomes: $p = 0.18$). Box's M test, used to assess homogeneity of covariance matrices, yielded a non-significant result (Box's $M = 18.75$, $F(10,36) = 1.62$, $p = 0.10$), indicating that the assumption of homogeneity of covariance matrices was met, thus validating the use of one-way ANOVA. Furthermore, the study design, which involved independent assignment of participants to groups, ensured the assumption of independence was satisfied.

3. Results

Table 1 presents the pretest Ms and SDs for motivation and language learning outcomes across the three groups.

As shown in Table 1, pretest motivation levels were comparable across groups, with the NFG ($M = 59.55$, $SD = 12.08$) scoring slightly higher than the DFG ($M = 58.21$, $SD = 11.07$) and IFG ($M = 57.29$, $SD = 11.65$). Similarly, language learning outcomes at

Table 1
Descriptive statistics results for pretest motivation and language learning outcomes.

		N	Mean	Std. Deviation	Std. Error
Motivation	IFG	31	57.29	11.65	2.09
	DFG	33	58.21	11.07	1.92
	NFG	34	59.55	12.08	2.07
	Total	98	58.38	11.53	1.16
Language learning outcomes	IFG	31	23.03	6.58	1.18
	DFG	33	23.90	6.88	1.19
	NFG	34	24.17	5.71	.97
	Total	98	23.72	6.35	.64

baseline were closely aligned, with the NFG again exhibiting a marginal advantage ($M = 24.17$, $SD = 5.71$) over the DFG ($M = 23.90$, $SD = 6.88$) and IFG ($M = 23.03$, $SD = 6.58$). To statistically verify these initial similarities, a one-way ANOVA was conducted (Table 2).

The ANOVA results (Table 2) confirmed no significant differences in pretest motivation ($F(2, 95) = 0.31$, $p = .73$) or language learning outcomes ($F(2, 95) = 0.28$, $p = .75$) across groups. This indicates that randomization successfully produced equivalent groups prior to the intervention, ensuring that any posttest differences could be reasonably attributed to the experimental conditions rather than pre-existing disparities.

Following the intervention period, we analyzed posttest scores to evaluate the differential effects of feedback timing. Table 3 presents the descriptive statistics for motivation and language learning outcomes across the three experimental conditions.

The posttest results revealed notable differences between groups. Both feedback groups (IFG and DFG) demonstrated substantially higher motivation scores compared to the NFG, with the DFG showing the highest average motivation ($M = 83.72$, $SD = 23.38$). A similar pattern emerged for language learning outcomes, where the DFG achieved the highest mean score ($M = 31.45$, $SD = 6.10$), followed by the IFG ($M = 30.16$, $SD = 6.53$). While the NFG showed improvement from pretest levels, their posttest performance in both motivation ($M = 63.08$) and language learning outcomes ($M = 26.38$) remained below the feedback groups. To determine whether these differences were statistically significant, we conducted one-way ANOVAs for each dependent variable (Table 4).

As seen in Table 4, the results confirmed statistically significant differences between groups for both motivation ($F(2, 95) = 10.79$, $p < .001$) and language learning outcomes ($F(2, 95) = 6.70$, $p = .002$). These findings suggest that feedback timing significantly influenced learners' motivational levels and language acquisition, with both immediate and delayed feedback proving more effective than no feedback. To identify specific group differences, Scheffé post hoc tests were conducted following significant ANOVA results. Table 5 presents the pairwise comparisons for both dependent variables.

As seen in Table 5, the Scheffé tests revealed three key findings. First, for motivation, both feedback groups significantly outperformed the NFG (IFG-NFG: $MD = 17.13$, $p = .002$; DFG-NFG: $MD = 20.63$, $p < .001$), with the DFG showing marginally higher (though not statistically significant) motivation than the IFG ($MD = 3.50$, $p = .77$). Second, language learning outcomes followed a similar pattern, with both feedback groups demonstrating superior performance to the NFG (IFG-NFG: $MD = 3.77$, $p = .04$; DFG-NFG: $MD = 5.07$, $p = .002$). Third, the DFG's advantage over the IFG in language learning outcomes, while notable (1.29 points), did not reach statistical significance ($p = .68$). These results advocate that while both feedback approaches effectively enhanced motivation and language learning outcomes compared to no feedback.

4. Discussion of Findings

The first research question examined the impact of immediate and delayed feedback timing on student motivation. The current study revealed that both IFG and DFG exhibited significantly higher motivation compared to the control group. This aligns with Fisher et al. (2025), who demonstrated that timely formative feedback positively influences motivational tendencies among university students. Similarly, Muis et al. (2015) highlighted that technology-mediated immediate feedback enhances students' attitudes, emotions, and engagement, all of which contribute to heightened motivation. These consistent findings suggest that the provision of timely feedback, irrespective of its immediacy, serves as a potent catalyst for stimulating learning engagement and fostering a more motivated learning environment.

The second research question focused on the effect of feedback timing on language learning outcomes. The results indicated that both experimental groups achieved significantly better learning outcomes on the posttest. This finding is corroborated by Lu et al. (2023), who established that immediate text-based feedback positively impacts foreign language learners' performance in online assignments. Furthermore, Kuklick (2024) demonstrated that both automated and immediate feedback significantly influence language learning outcomes. Hakim et al. (2022) also found that immediate feedback mechanisms are associated with increased learning performance among students.¹ These studies collectively reinforce the notion that feedback, whether delivered immediately or with a delay, plays a crucial role in enhancing language learning outcomes. Additionally, the study's results are consistent with Taxipulati and Lu (2021) who found that the timing of the feedback affected students' achievement within multimedia learning contexts.

Expanding beyond language learning, Kim and Arbel (2019) demonstrated that both immediate and delayed auditory feedback are beneficial for declarative learning, suggesting that the timing of feedback, regardless of immediacy, aids learners. Similarly, Almalki and Elfeky (2022) found that both immediate and delayed feedback modalities enhance higher-order thinking skills in virtual classes. Xie and Zhang (2024) further emphasize the importance of online peer feedback design and quality, thereby extending the understanding of feedback's substantial role in the learning process. Attali and Van Der Kleij (2017) also found that timing and elaboration of feedback improved computer-based problem solving in mathematics.

Table 2
ANOVA results for pretest motivation and language learning outcomes.

		Sum of Squares	df	Mean Square	F	Sig.
Motivation	Between Groups	84.98	2	42.49	.31	.73
	Within Groups	12,816.28	95	134.90		
	Total	12,901.26	97			
Language learning outcomes	Between Groups	22.92	2	11.46	.28	.75
	Within Groups	3894.63	95	40.99		
	Total	3917.56	97			

Table 3

Descriptive statistics results for posttest motivation and language learning outcomes.

		N	Mean	Std. Deviation	Std. Error
Motivation	IFG	31	80.22	21.02	3.77
	DFG	33	83.72	23.38	4.07
	NFG	34	63.08	12.50	2.14
	Total	98	75.45	21.31	2.15
Language learning outcomes	IFG	31	30.16	6.53	1.17
	DFG	33	31.45	6.10	1.06
	NFG	34	26.38	5.00	.85
	Total	98	29.28	6.23	.62

Table 4

ANOVA results for posttest motivation and language learning Outcomes.

		Sum of Squares	df	Mean Square	F	Sig.
Motivation	Between Groups	8163.63	2	4081.81	10.79	.00
	Within Groups	35,918.70	95	378.09		
	Total	44,082.33	97			
Language learning outcomes	Between Groups	465.59	2	232.79	6.70	.00
	Within Groups	3300.40	95	34.74		
	Total	3766.00	97			

Table 5

Posttest multiple comparisons (Scheffé) for motivation and language learning outcomes.

Dependent Variable	(I) groups	(J) groups	Mean Difference (I-J)	Std. Error	Sig.
Motivation	IFG	DFG	-3.50	4.86	.77
		NFG	17.13	4.82	.00
		DFG	3.50	4.86	.77
	DFG	NFG	20.63	4.75	.00
		IFG	-17.13	4.82	.00
		DFG	-20.63	4.75	.00
Language learning outcomes	IFG	DFG	-1.29	1.47	.68
		NFG	3.77	1.46	.04
		DFG	1.29	1.47	.68
	DFG	NFG	5.07	1.44	.00
		IFG	-3.77	1.46	.04
		DFG	-5.07	1.44	.00

The Expectancy-Value Theory (EVT) (Eccles & Wigfield, 2002) and Constructivism (Vygotsky, 1978) provide compelling explanations for the observed motivational benefits. According to EVT, learners' motivation is shaped by their expectations of success and the perceived value of the task. In this study, immediate feedback may have reinforced learners' self-efficacy by offering rapid validation of their efforts, while delayed feedback encouraged deeper reflection on the intrinsic value of improvement. This dual mechanism aligns with prior research suggesting that different feedback timings cater to distinct motivational drivers (Lipnevich & Panadero, 2021). Similarly, Vygotsky's Zone of Proximal Development (ZPD) underscores the role of scaffolded support in knowledge construction. Immediate feedback likely prevented early misconceptions, whereas delayed feedback allowed for more autonomous cognitive processing, facilitating progression through the ZPD. This interpretation is consistent with studies demonstrating that delayed feedback can promote higher-order thinking (Hattie & Timperley, 2007).

The findings are further supported by FIT (Kluger & DeNisi, 1996) and Krashen's Input Hypothesis (1982). FIT posits that feedback directs attention toward task improvement rather than self-evaluation, thereby enhancing self-regulation. In this study, both immediate and delayed feedback functioned as interventions that helped learners identify and correct errors, though through different cognitive pathways. Immediate feedback likely facilitated quick adjustments, while delayed feedback encouraged metacognitive reflection, contributing to sustained performance gains (Juffer et al., 2017). The Input Hypothesis complements this perspective by suggesting that feedback, whether immediate or delayed, serves as comprehensible input, enabling learners to refine their linguistic output. Immediate feedback may have aided real-time accuracy, whereas delayed feedback promoted consolidation, aligning with evidence that spacing corrections can improve retention (Hubackova, 2016).

Self-Determination Theory (SDT) (Deci & Ryan, 2000) offers additional insights into the motivational effects observed. SDT emphasizes that learners thrive when their needs for competence, autonomy, and relatedness are met. Immediate feedback may have bolstered competence by clarifying errors promptly, while delayed feedback supported autonomy by allowing learners time to self-assess before receiving corrections (Dainer-Best & Rubin, 2024). The structured nature of both feedback types may have also fostered relatedness by creating a sense of guided progress, which is critical in online learning environments (Ryan & Deci, 2020).

From a cognitive perspective, Cognitive Load Theory (CLT) (Sweller, 1988) helps explain why delayed feedback may lead to deeper learning. CLT suggests that learners have limited working memory capacity, and immediate feedback—if too frequent—can overwhelm them. Delayed feedback, by contrast, provides a reflective pause, reducing cognitive load and enabling more effective integration of corrections (Arbel et al., 2017). This may explain why both feedback timings were beneficial but potentially operated through different mechanisms: immediate feedback for quick error resolution and delayed feedback for long-term retention.

Collectively, these theoretical frameworks suggest that immediate and delayed feedback serve complementary yet distinct roles in online learning. While immediate feedback enhances motivation through rapid reinforcement, delayed feedback supports deeper cognitive engagement. The interplay between these mechanisms likely contributed to the improved outcomes observed in this study. Future research could further disentangle the specific conditions under which each feedback timing is most effective, particularly in diverse learning contexts.

5. Conclusions

This study aimed to examine the effects of immediate versus delayed feedback timing on Chinese EFL learners' motivation and language learning outcomes within online education, framed by FIT. The findings revealed that both immediate and delayed feedback significantly enhanced learners' motivation compared to the control group. Additionally, both experimental groups demonstrated improved language learning outcomes on the posttest, suggesting that structured feedback—regardless of timing—positively influences performance. The results indicate that while immediate feedback facilitates real-time error correction, helping learners refine their language use promptly, delayed feedback encourages deeper reflection, supporting long-term retention. Importantly, no substantial difference was found between the two feedback timings, implying that both approaches are effective but may serve different cognitive and motivational functions. These findings underscore the critical role of feedback in language acquisition, whether through immediate reinforcement or delayed, reflective processing.

The study's findings provide actionable insights relevant to EFL learners, teachers, and curriculum designers. Strategic integration of immediate and delayed feedback mechanisms significantly enhances the language learning experience. Immediate feedback delivers real-time corrective guidance, bolstering confidence and self-efficacy through clear, actionable steps for improvement. This immediate reinforcement is particularly valuable during interactive tasks, enabling learners to adjust their language use instantaneously. Conversely, delayed feedback encourages deeper cognitive processing and reflection, fostering long-term retention and understanding. As Molin et al. (2021) suggest, this reflective process cultivates problem-solving skills and promotes learner autonomy. Therefore, EFL learners should actively seek opportunities to engage with both forms of feedback, recognizing their complementary roles in language acquisition.

The study underlines the importance of adopting a flexible and balanced feedback approach. Incorporating both immediate and delayed feedback into teaching practices caters to diverse learning needs and optimizes student outcomes. Immediate feedback, as highlighted by Gan et al. (2021), is particularly effective during in-class activities, preventing the internalization of errors and maintaining lesson flow. Delayed feedback, on the other hand, is well-suited for written assignments and homework, providing learners with time to engage in critical self-evaluation before receiving external input. Crucially, educators must remain cognizant of learners' cognitive load, ensuring that feedback is delivered in manageable quantities and with clear, concise language. Implementing these strategies creates a supportive learning environment that fosters motivation, enhances language proficiency, and cultivates self-regulatory skills. Moreover, curriculum designers can utilize these findings to create materials and online learning environments that effectively integrate both types of feedback.

This study is not without limitations that should be acknowledged. Firstly, the sample size, comprising 98 participants, presents a constraint on the generalizability of the findings. Future research could enhance external validity by employing larger, more diverse samples, thereby capturing a broader spectrum of learner experiences. Secondly, the study did not explore the potential influence of gender on feedback efficacy. Given that gender-related differences in learning preferences and feedback receptivity have been observed in other educational contexts, subsequent investigations should incorporate gender as a key variable. This would allow for a more nuanced understanding of how feedback timing interacts with gender to influence learning outcomes.

Thirdly, the homogeneity of the participant pool, drawn exclusively from an intermediate-level English language institute in Xi'an, China, limits the applicability of the results to other proficiency levels and cultural backgrounds. To address this, future studies could adopt a cross-cultural comparative design, examining the effects of feedback timing across diverse cultural contexts and proficiency levels. Such an approach would provide valuable insights into the universality and cultural specificity of feedback effectiveness. Fourthly, the study did not account for individual differences in learning styles and prior experiences with feedback. These factors may significantly influence how learners respond to immediate or delayed feedback. Future research could employ mixed-methods approaches, incorporating qualitative data to explore individual learner experiences and preferences. Moreover, novel investigations could explore the interaction between learner personality traits (e.g., conscientiousness, openness to experience) and feedback timing, thus deepening our understanding of individual variation. Finally, the study's focus on short-term learning outcomes necessitates further investigation into the long-term effects of feedback timing on language learning retention. Innovative longitudinal studies could track learner progress over extended periods, examining how different feedback timings influence long-term retention and application of language skills. Furthermore, research could explore adaptive feedback systems that tailor feedback timing and content to individual learner needs and progress, potentially revolutionizing personalized learning in online EFL environments. This approach would allow for a more dynamic and responsive learning experience, optimizing both motivation and learning outcomes.

CRedit authorship contribution statement

Yang Qun: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

Data availability

Data will be made available on request.

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